

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-II	20.00	23.68
2	Module-V	20.00	21.05
3	Module-III	20.00	19.74
4	Module-IV	20.00	19.74
5	Module-I	20.00	15.79

Weightage representation okay : Weightage representation exceeded more than 5% :Weightage representation exceeded more than 10%

CSIT617

[ET]

Enroll. No. - _____

END SEMESTER EXAMINATION : April-May, 2026

Python for Data Science

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

Q1.

(a)

How do you create and store strings in Python? List any five string functions.

(3)

Delete

(b)

Explain the use of numpy for creating and managing arrays.

(3)

Delete

Q2.

(a)

What is a Lambda function? Provide a syntax example.

(3)

Delete

(b)

Describe the try-except-else-finally block structure.

(3)

Delete

Q3.

Compare and contrast Data Science and Data Analytics.

(6)

Delete

Q4.

(a)

What is the difference between Normal and Exponential continuous distributions?

(3)

Delete

(b)

Explain the role of Subplots and their properties in matplotlib for data visualization.

(3)

Delete

Q5.

(a)

Define Random Variables in the context of inferential statistics.

(3)

Delete

(b)

Explain the hierarchy of exceptions in Python.

(3)

Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

Q6.

(a)

Explain the commonly used functions on Dictionaries and Tuples.

(5)

Delete

(b)

How do you use the math module for mathematical operations on lists?

(5)

Delete

Q7.

(a)

Explain how to explore data statistics using the stats package from SciPy.

(5)

Delete

(b)

Describe the process of managing data with a relational database (MySQL) via Python.

(5)

Delete

Q8.

(a)

Discuss the steps involved in Data Cleaning and removing duplicates.

(5)

Delete

(b)

Explain the concept of a Cumulative Distribution Function (CDF).

(5)

Delete

Section - C : Compulsory question

Q9.

(a)

Explain the Scope and Lifetime of variables in Python functions.

(4)

Delete

(b)

Discuss the measurement scales of data (Nominal, Ordinal, Interval, Ratio).

(4)

Delete

(c)

Describe the Standard Normal Distribution and its properties.

(4)

Delete

(d)

Compare basic visualization in matplotlib with advanced visualization in Seaborn.

(4)

Delete